

A general framework for using two-stage IPD-MA to predict individualized treatment effects

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- data from single RCTs are not suitable
- we need multiple trials to investigate effect modification and make predictions

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Two-stage approach

1. analyses are performed **separately** for each study
2. study-specific results are pooled and synthesized

Aims of the study

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Aims of the study

1. present a **general** two-stage IPD meta-analysis **framework** for predicting **ITE**
 - **Within-trial flexibility** - any predictive model for ITE can be employed in first stage
 - **Between-trial adaptability** - different models can be used across studies
2. evaluate performances with a **simulation study** for a continuous outcome

Framework pipeline: a two-stage approach (I)

First stage: fit study-specific models.

This can be done with **any type of model**:

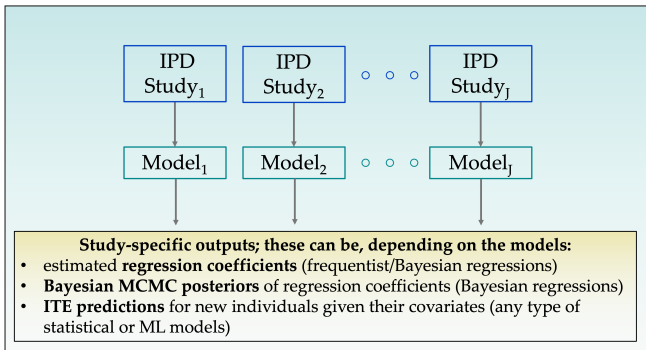
- Frequentist regression models (OLS, LASSO, Ridge)
- Bayesian regression models
- Machine learning (XGBoost, Causal Forest)
- Meta-learners (R- and X-)

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Framework pipeline: a two-stage approach (II)

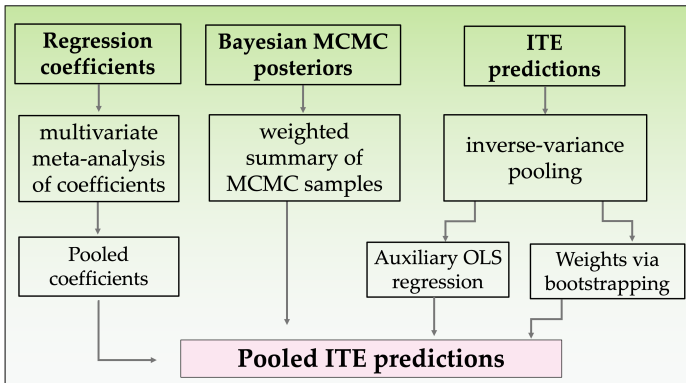
Second stage: pooling of study-specific results

- We used different pooling methods to yield ITE predictions
- Applicable second stage methods **depends on the choice of first-stage models**

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- study-specific sample size (n)
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Performance was measured with Median Absolute Error (MAE), Mean Bias and R^2

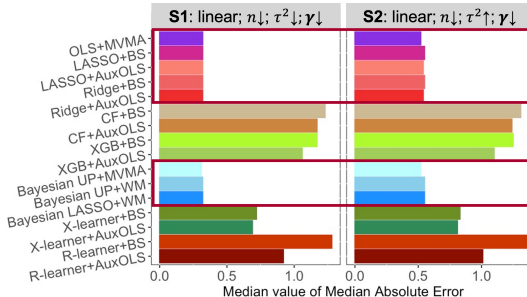
Simulation results: Models comparison (I)

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- In scenarios with linear interactions and small samples, **regression-based models** in the first stage consistently outperformed machine learning

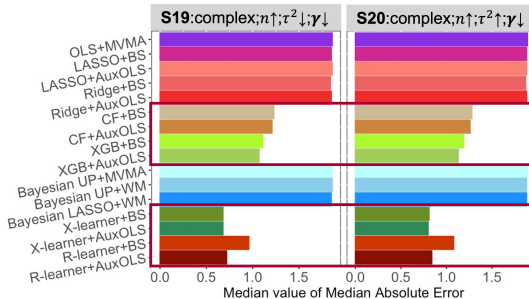


Simulation results: Models comparison (II)

- With complex interactions and large samples, **machine learning** approaches performed better than linear regressions
- In particular, **weighting by OLS inverse-variance** in the second stage performed better than bootstrapping

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- Simulations showed how performance depended on the way data were generated, for example:
 - linear models work well with small samples
 - with larger samples and complex effect modifications, machine learning lead to more accurate predictions
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 - linear models work well with small samples
 - with larger samples and complex effect modifications, machine learning lead to more accurate predictions
 - meta-learners (R- and X-) with XGBoost performed even better
- However, data generation mechanisms are not known in real world applications. Specification of the framework and model selection should be based on proper validation

Thank you for your attention!

Publication

A general framework for using two-stage meta-analysis with individual participant data to predict individualized treatment effects

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